Statistical Data Analysis

Lecture 12: Linear regression analysis, part 2

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Periods 4 & 5

Where Are We?

- Chapter 7: Analysis of categorical data
  - Still today: deadline of Assignment 6 & Quiz 6. Discussion next week
- Chapter 8: Linear regression analysis
  - Section 8.1 – The multiple linear regression model
  - Section 8.2 – Diagnostics
  - Section 8.3 – Collinearity

Recap – multiple linear regression

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- Linaer regression equation (scalar)

$$Y_i = \beta_0 + x_{i1}\beta_1 + \cdots + x_{ip}\beta_p + e_i$$

- Linear regression equation (vectorized)

$$\boldsymbol{Y} = \boldsymbol{X}\beta + \boldsymbol{e}$$

- Assumptions on errors

$$\mathbb{E}(e_i) = 0 \qquad \mathbb{E}(e_i e_j) = \begin{cases} \sigma^2, & i = j, \\ 0, & i \neq j \end{cases}$$

- Further assumptions
  - Full-rank design matrix, i.e. $\text{rank}(\boldsymbol{X}) = p + 1$
  - $e_i \sim N(0, \sigma^2)$, so that $Y \mid \boldsymbol{X} = \boldsymbol{x} \sim N\left(\beta_0 + x_1\beta_1 + \cdots + x_p\beta_p, \sigma^2\right)$

Variable Selection – Coefficient of determination

- Given $\boldsymbol{X}$, we estimate Y by $\hat{y}_i = f_{\hat{\beta}}(\boldsymbol{x}_i) = \boldsymbol{X}\hat{\beta}$.
- For the total variance in the data, we have

$$\mathrm{SSY} = \mathrm{RSS} + \mathrm{SS}_{\mathrm{reg}}$$

Variance caused by the lineair model $\sum(\hat{y}_i - \bar{y})^2$

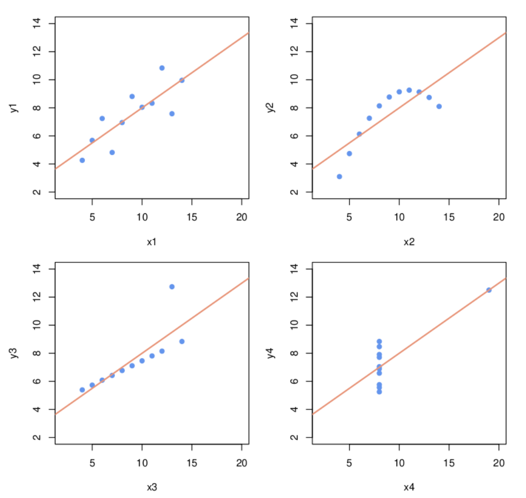
Variance caused by errors in the regression $\sum(y_i - \hat{y}_i)^2$

Total variance $\sum(y_i - \bar{y})^2$

- RSS is the unexplained variance
- $\mathrm{SS}_{\mathrm{reg}}$ is the explained variantie
- **Coefficient of determination:** $\mathcal{R}^2 = \frac{\mathrm{SS}_{\mathrm{reg}}}{\mathrm{SSY}} = 1 - \frac{\mathrm{RSS}}{\mathrm{SSY}}$

Diagnostics

Diagnostics – Motivation



For each data set in the Anscombe's quartet, fitting a simple linear regression model via OLS results in $\hat{\beta}_0 = 3$, $\hat{\beta}_1 = 0.5$, $\hat{\sigma}^2 = 1.5$ and $\mathcal{R}^2 = 0.67 \ldots$ But is this model appropriate for all four data sets? Why/why not?

Diagnostics – Potential Problems

- When estimating a model and devising testing procedures, we do so under some assumptions
  - For linear regression, we assumed linearity and error normality
- To investigate the plausability of such assumptions and the overall quality of the model, we resort to diagnostics
  - Diagnostic tools: graphical and numerical checks
  - Potential problems: non-linearity, non-constant error variance, outliers, leverage points, (multi)collinearity,. . .

Scatterplots for Diagnostics

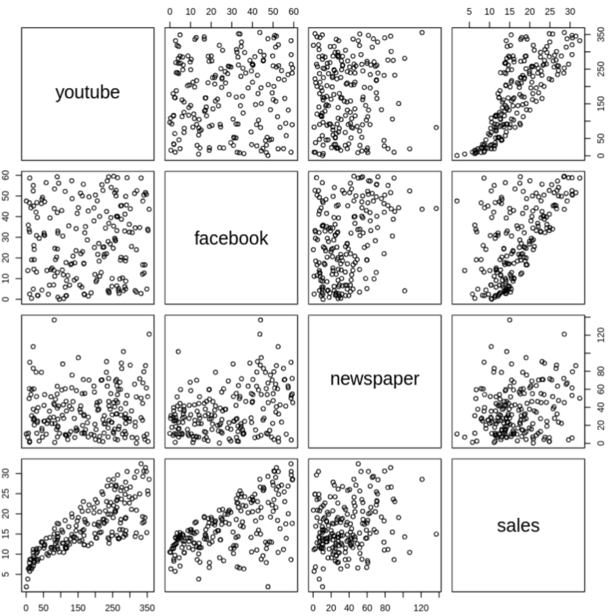
Response vs. Predictors & Pairs of Predictors

- Scatterplots of response against each predictor
  - They help to get an overall picture and to identify outlying values
- Scatterplots for each pair of explanatory variables
  - They help visualize potential relationships between predictors

```r
### Impact of advertising media on sales (in thousands of dollars)
library("datarium"); data(marketing)
mod <- lm(sales ~ youtube + facebook + newspaper, data = marketing)
summary(mod)
```

```
##
## Call:
## lm(formula = sales ~ youtube + facebook + newspaper, data = marketing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.5932  -1.0690   0.2902   1.4272   3.3951
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.526667   0.374290   9.422   <2e-16 ***
## youtube      0.045765   0.001395  32.809   <2e-16 ***
## facebook     0.188530   0.008611  21.893   <2e-16 ***
## newspaper   -0.001037   0.005871  -0.177     0.86
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.023 on 196 degrees of freedom
## Multiple R-squared:  0.8972, Adjusted R-squared:  0.8956
## F-statistic: 570.3 on 3 and 196 DF,  p-value: < 2.2e-16
```

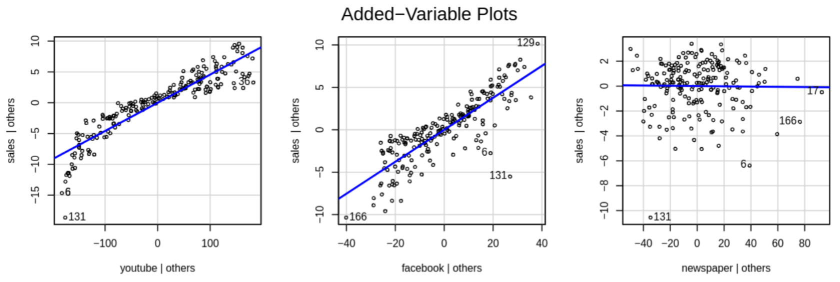
Response vs. Predictors & Pairs of Predictors - Using `pairs`



Added Variable Plots – Do we add information by including the variable X_k?

- Added variable plot for X_k
  - Scatterplot of residuals $R_Y(X_{-k})$ (from the regression of Y on all predictors except X_k) against residuals $R_{X_k}(X_{-k})$ (from the regression of X_k on all other predictors)

```r
library("car"); avPlots(mod, layout = c(1, 3))
```



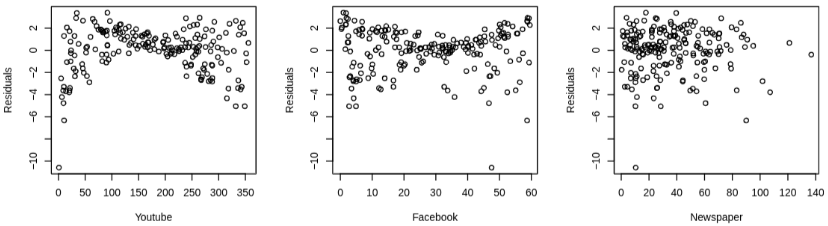
Added variable plots– Do we add information by including the variable X_k?

- Added variable plot for X_k
  - x-axis: everything from X_k not explained by other variables
  - y-axis: everything from Y not explained by other variables.
  - Shows the relationship between Y and X_k after correcting for possible relationships between X_k and the other predictors
  - Noticeable linear (but non-flat) pattern $\rightsquigarrow$ X_k should also be included in the model
  - In that case, X_k adds information about Y, which is not already in the other variables

Residuals vs. (Potential) Predictors – homoscedasticity?

- Scatterplots of residuals against each (included) predictor
  - Curvilinear pattern ⇝ higher-order terms may need to be included
  - Systematic pattern in the spread of the points ⇝ error variance may not be constant (so-called heteroscedasticity)
- Scatterplots of residuals against each potential predictor
  - Noticeable linear (but non-flat) pattern ⇝ variable should be included in the model

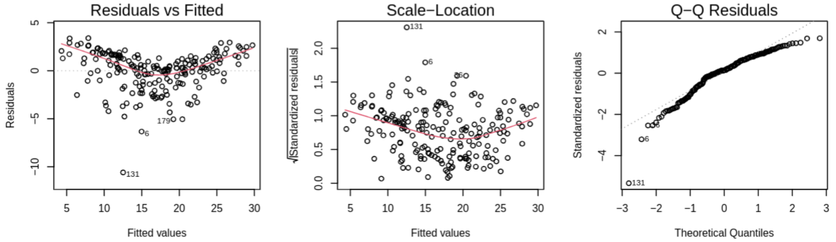
```
par(mfrow = c(1, 3))
plot(mod$residuals ~ marketing$youtube, xlab = "Youtube", ylab = "Residuals")
plot(mod$residuals ~ marketing$facebook, xlab = "Facebook", ylab = "Residuals")
plot(mod$residuals ~ marketing$newspaper, xlab = "Newspaper", ylab = "Residuals")
```



Residuals vs. Fitted & Normal QQ-Plot of Residuals

- Scatterplots of residuals against fitted (i.e., predicted) values
  - Systematic pattern (e.g. funnel shape) ⇝ heteroscedasticity
  - Possible remedies: variable transformation, weighted linear regression
  - Strong nonlinear relationship ⇝ linear model not appropriate
  - Possible remedies: variable transformation, nonlinear model
- Normal QQ-plot of residuals
  - Strong deviations from QQ-line ⇝ normality assumption not satisfied

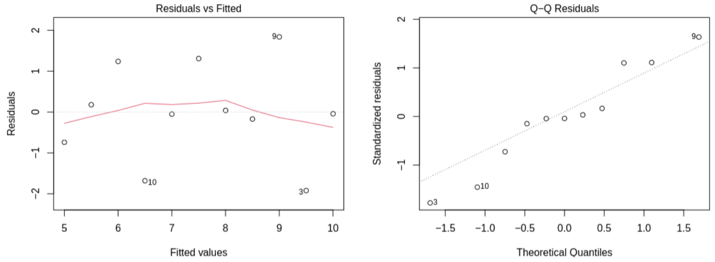
```r
par(mfrow = c(1, 3))
plot(mod, which = 1); plot(mod, which = 3); plot(mod, which = 2)
```



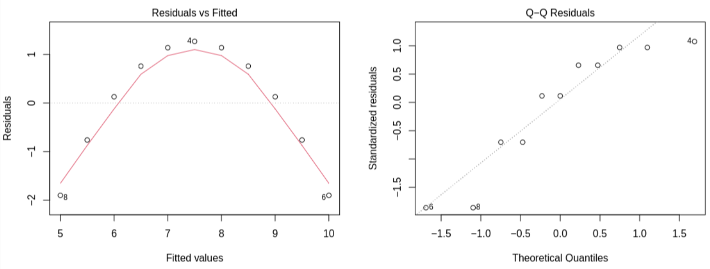
Are there any clear issues with the model fitted in this example?

Residuals vs. Fitted & QQ-Plot of Residuals – Anscombe 1 & 2

Anscombe's Dataset 1

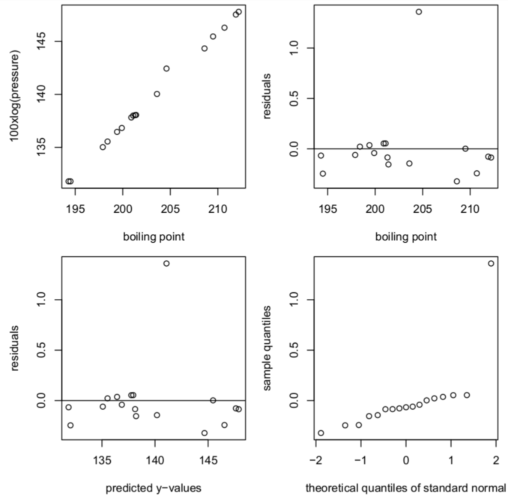


Anscombe's Dataset 2



Outliers

- Outlier: observation with extremely high or low response value compared to what is to be expected under the model



Detecting outliers – Mean Shift Outlier Model

- Does the k^{th} point deviate significantly from the others?
- Mean shift outlier model: $Y_i = \begin{cases} x_i^T\beta + e_i & \text{if } i \neq k \\ x_i^T\beta + \delta + e_i & \text{if } i = k \end{cases}$
  - Alternative notation: $\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{u}\delta + \boldsymbol{e}$ with $u_i = 0$ for $i \neq k$ and $u_k = 1$
  - Idea: adapt the original model so it can fit the k^{th} point
  - The expectation of the k^{th} response is shifted by δ
  - Use a t-test to test $H_0 : \delta = 0$ vs. $H_1 : \delta \neq 0$: if H_0 rejected $\rightsquigarrow$ outlier

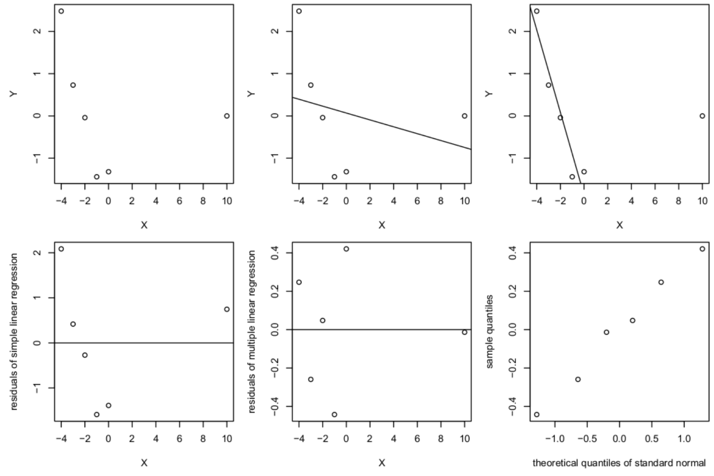
```r
library("MASS") # Contains Forbes data
# Create response y by tranforming pressure as in Syllabus
forbes$y <- 100 * log(forbes$pres, base = 10)
# Create fictional 'explanatory' variable u
forbes$u <- c(rep(0, 11), 1, rep(0, 5))
# Fit mean shift outlier model
summary(lm(y ~ bp + u, data = forbes))$coefficients
```

```
##               Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) -41.3346826 1.003312257 -41.19822 5.155989e-16
## bp            0.8911095 0.004944098 180.23703 5.767389e-25
## u             1.4528305 0.117411265  12.37386 6.302461e-09
```

Leverage Points

Leverage Points – Definition – Huber Fictive Data

- Leverage point: observation with extremely high or low value of explanatory variable
  - Note that checking for large residuals does not suffice in this case



For details and explanations about the above plots, see Example 8.8 in the Syllabus (in particular, the caption of Figure 8.6)

Leverage Points – Detection – Hat Matrix

- Write the predicted response has $\hat{\boldsymbol{Y}} = \boldsymbol{X}\hat{\beta} = \boldsymbol{X}\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T\boldsymbol{Y} =: \boldsymbol{H}\boldsymbol{Y}$
  - Hat matrix $\boldsymbol{H} := \boldsymbol{X}\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T$ 'transforms' $\boldsymbol{Y}$ into predictions $\hat{\boldsymbol{Y}}$ ('y-hat')
  - Its diagonal elements $h_{ii} = x_i^T\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}x_i \in [0, 1]$, $i = 1, \ldots, n$ are useful to investigate the influence of extreme values

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- From Syllabus: $\mathrm{Var}(R_i) = \sigma^2(1 - h_{ii})$
  - $h_{ii} \approx 1 \rightsquigarrow \mathrm{Var}(R_i)$ small, fit is pulled towards Y_i
  - Typical condition: $h_{ii} > 2 \cdot (p+1)/n \rightsquigarrow i^{\text{th}}$ observation is a leverage point
  - Depending on the value of Y_i, potential influence

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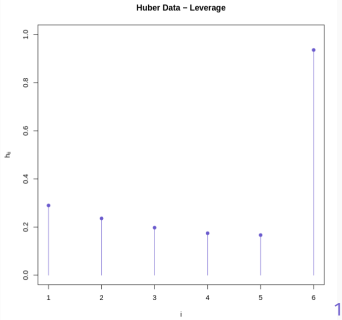
```
huber <- data.frame(x = c(-4:0, 10),
                    y = c(2.48, 0.73, -0.04, -1.44, -1.32, 0))
round(hatvalues(lm(y ~ x, data = huber)), 3)
```

```
##     1     2     3     4     5     6
## 0.290 0.236 0.197 0.174 0.167 0.936
```



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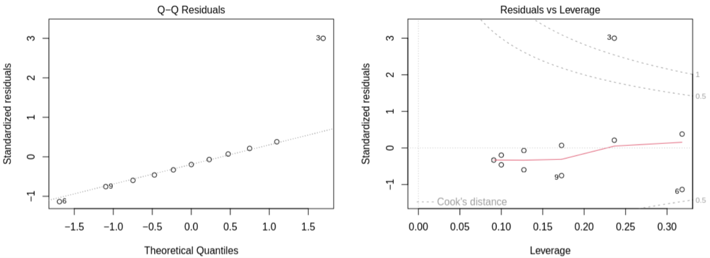
- Outliers and leverage points are not necessarily influential but have the potential to be. How to judge?
- Idea: fit model with and without i^{th} data point
  - Big change in fitting $\rightsquigarrow$ i^{th} data point is an influence point
- Cook's distance for i^{th} data point: $D_i = \frac{(\hat{\mathbf{Y}}_{(i)} - \hat{\mathbf{Y}})^T(\hat{\mathbf{Y}}_{(i)} - \hat{\mathbf{Y}})}{(p+1)\hat{\sigma}^2}$
  - $\hat{\mathbf{Y}}_{(i)}$ denotes predictions by the model without the i^{th} data point
  - It can be shown (with standardized residual r_i):

$$D_i = \frac{r_i^2}{p}\frac{h_{ii}}{1 - h_{ii}}$$

  - Rule of thumb: $D_i \approx 1$ or $D_i > 1 \rightsquigarrow$ influence point

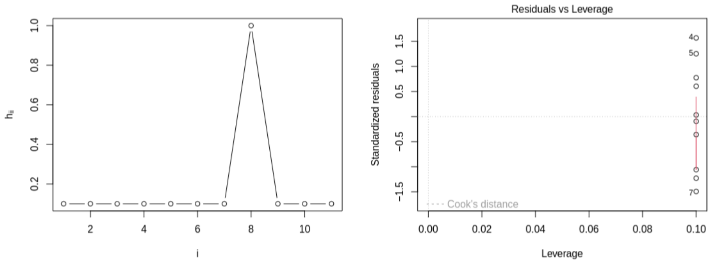
Outliers & Leverage Points – Anscombe 3 & 4

Anscombe's Dataset 3

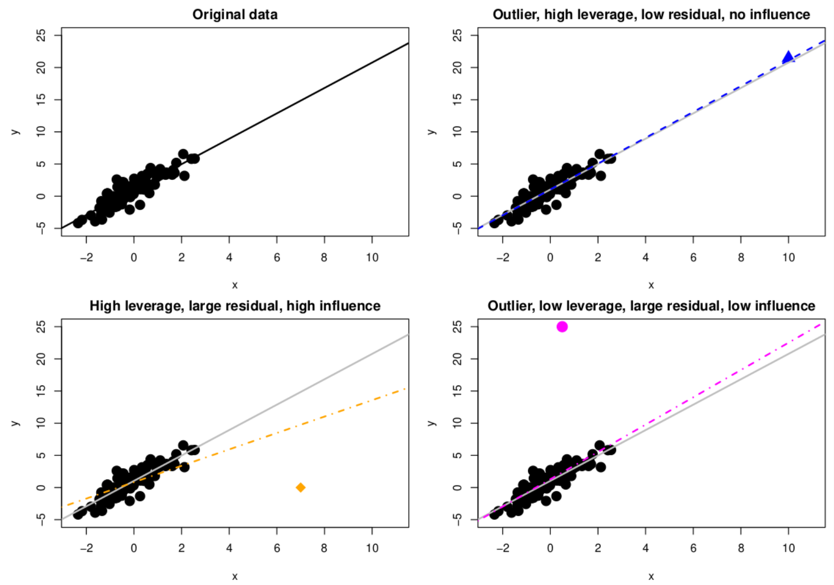


Anscombe's Dataset 4 (note the warning)

```
## Warning: not plotting observations with leverage one:
##   8
```



Outliers, Leverage & Influence



Summary & wrap-up

Summary

- Diagnostics
  - Added variable plots for variable selection
  - Residuals vs. predictors for homoscedasticity
  - Residuals vs fitted values and normal QQ-plots for normality of residuals.
- Outliers
  - Mean shift outlier model for detection
  - Leverage points: extremal points for an explanatory variable
    - Hat values
  - Influence points: points that heavily influence the model
    - Cook's distance

What now?

- To do for now
  - Deadline Assignment 6: today at 23:59
  - Start Assignment 7
  - All material covered after next week
  - **Deadline:** May 19 at 23:59
- After that
  - Last lecture next week
  - Discussion of A7 and old exam on May 20.